Writing an IT Project Report (BTech and MTech)

First, check your department or university's format guidelines (for example AKTU, MNNIT, or your institute's thesis manual). Those override anything generic, including margins, fonts, and the order of pages.

Standard structure

Front matter

1. Title page: title, your name and roll number, guide, department, institute, month and year
2. Certificate (signed by guide and HOD)
3. Declaration (the work is your own)
4. Acknowledgement
5. Abstract (150-300 words): problem, approach, key result
6. Table of contents, list of figures, list of tables, list of abbreviations

Main chapters

1. **Introduction**: background, problem statement, motivation, objectives, scope, and report organization.
2. **Literature Review**: what existing work has been done, compared critically (a comparison table works well), ending with the research gap your project addresses.
3. **System Analysis and Requirements**: functional and non-functional requirements, feasibility study, hardware/software requirements, and the tools and technologies chosen, with justification.
4. **System Design**: architecture diagram, data flow diagrams, ER diagram, UML (use case, class, sequence), database schema, algorithm or flowchart, UI wireframes.
5. **Implementation**: modules explained one by one, key code snippets (not entire files), the algorithms or models used, and challenges you faced and how you solved them.
6. **Testing and Results**: test plan, test cases with expected vs. actual output, screenshots, performance metrics (accuracy, response time, precision/recall, and so on), and comparison with existing systems or baselines.
7. **Conclusion and Future Scope**: what you achieved against your objectives, limitations, and realistic extensions.

Back matter

- References (IEEE format is standard in IT)
- Appendices: full code, questionnaires, datasets, user manual, plagiarism report if required

BTech vs. MTech

	BTech	MTech
Focus	Building a working system,	Research contribution, novelty, analysis

	BTech	MTech
	applying known techniques	
Literature review	10-20 references, mostly background	40-60+ references, mostly recent journal and conference papers
Evaluation	Functional testing, screenshots	Experiments, baselines, statistical comparison, ablation or sensitivity analysis
Outcome	Working prototype	Often a publishable paper, with a clearly stated contribution
Length	About 50-80 pages	About 80-150 pages

For MTech, state your contributions explicitly, for example "The main contributions of this work are…", and make sure your results chapter proves each one.

Formatting norms (typical, so verify)

- Times New Roman 12 pt, 1.5 line spacing, justified text
- Headings: 14 pt bold for chapters, 12-13 pt bold for sections
- Margins: left 1.5", others 1"
- Numbered chapters, sections (3.1, 3.2), figures (Fig. 3.2) and tables (Table 4.1), each with a caption and a mention in the text
- Roman numerals for front matter, Arabic for the main body

Writing tips

- **Write as you build.** Draft the literature review and design chapters early. Don't leave everything for the last month.
- **Use past tense for what you did** ("The system was implemented using…") and present tense for general facts.
- **Avoid "I".** Use passive voice or "we", depending on your guide's preference.
- **Every figure and table must be referenced and explained** in the text.
- **Show evidence.** Back claims with results, graphs, or citations.
- **Keep code in the appendix**, with only the important logic in the main body.
- **Cite properly.** Plagiarism checks (Turnitin and similar) are common. Paraphrase in your own words and cite every source.

Common mistakes

- A vague problem statement with objectives that don't match the results
- A literature review that only lists papers without comparing them
- Pasting screenshots with no explanation
- A conclusion that repeats the abstract without addressing the objectives
- Inconsistent formatting, broken numbering, and out-of-date table of contents
- Missing or inconsistently formatted references

Quick workflow

1. Get your guide's approved outline.
2. Draft the chapters in the order you work: introduction → literature review → design → implementation → results.
3. Add figures and tables as you generate them.
4. Write the abstract and conclusion last.
5. Do a final pass for formatting, a plagiarism check, and proofreading, then get your guide's review before printing.